

Esercizio.

Let $(V, \epsilon) \models ZF^-$. Let M definable class, $(M, \epsilon) \subseteq (V, \epsilon)$. Let $A \subseteq M, R \subseteq A^2$ definable classes with parameters in M . Suppose:

1. M transitive, $M \models ZF^-$
2. R absolute
3. A absolute
4. $M \models R$ well founded , $M \models R$ set-like

Then the function defined internally in M as $rk_R : A \rightarrow Ord, b \mapsto \bigcup_{a \in pred_R(b)} S(rk_R(a))$ is absolute.

Dimostrazione.

First observe that since A, R are absolute and M is transitive we have that $pred_R$ is absolute. Now take as rule for the induction $\phi(x, y) : y = \bigcup \{x(b) + 1 : b \in dom(x)\}$ and note that it is $\Delta_1(ZF^-)$. One can now define

$$(x, y) \in rk_R \iff \exists w \left[w \text{ function} \wedge w(x) = y \wedge \right. \\ \left. \exists b \left(b \in A \wedge dom(w) = pred_R(b) \wedge \forall a \in pred_R(b) (\phi(w \upharpoonright pred_R(a), w(a))) \right) \right]$$

Note that the above formula is upward absolute since there are only existentials, absolute formulas and bounded quantification (in particular $\forall a \in pred_R(b)$ is a bounded quantification because $M \models R$ set-like).

Similarly one can define

$$(x, y) \in rk_R \iff \forall w \left[\left(w \text{ function} \wedge \right. \right. \\ \left. \left. \exists b \left(b \in A \wedge dom(w) = pred_R(b) \wedge \forall a \in pred_R(b) (\phi(w \upharpoonright pred_R(a), w(a))) \right) \right) \right. \\ \left. \rightarrow w(x) = y \right]$$

□